



SAT Math Concepts & Strategies from our August 2nd, 2026 session

- Picking Numbers
 - When you see variables, picking numbers can be a strategy to use to make it easier to evaluate the answer choices.
 - Pick easy numbers! -1 , $-\frac{1}{2}$, 0 , $\frac{1}{2}$, 1
 - Make sure the numbers you pick conform to the rules of the question
- FOILing and Factoring
 - FOIL
 - When you have two binomials, FOIL them to see if you can combine like terms with other elements in the question.
 - $(x - 3)(x + 4) = x^2 + 4x - 3x - 12 = x^2 + x - 12$
 - Factor
 - When you have a trinomial and you wish to turn it into two binomials.
 - Need two numbers that multiply to give you the third term (a constant) and that add up to give you the coefficient of the second term (the one that has x attached to it).
- Line concepts
 - Find the slope! The m in $y = mx + b$.
 - b is the y -intercept, which is when $x = 0$.
 - The change in y divided by the change in x
 - "Delta y divided by delta x "
 - **Parallel** lines has the same slope, and they never intersect.
 - **Perpendicular** lines have opposite, reciprocal slopes.
 - 2 is the opposite reciprocal of $-\frac{1}{2}$

- Triangles
 - Right triangles
 - 30-60-90 degree triangle
 - The side opposite the 30-degree angle is x , the side opposite the 60-degree angle is $x * \sqrt{3}$, and the side opposite the right angle is $2x$
 - 45-45-90 degree triangle
 - The sides opposite the 45-degree angles are each x and the side opposite the 90-degree angle is $x * \sqrt{2}$
 - 3-4-5 side proportions
 - 5-12-13 side proportions
- π radians = 180 degrees
- Scientific conversions
 - There are 3,600 seconds squared in a minute squared.
 - 1 km = 1,000 m
 - 62% less than $x = 0.38 * x$
- Undefined: a fraction is undefined when the denominator is 0.

Vocabulary we saw in the 8/2/26 session:

- Arbitrary: random, up to one's opinion (neutral)
 - The king arbitrarily decreed that the age for entering the army is to be 18 years old.
- Debased: made to be of poor quality, slipshod (-)
 - Texting has debased the English language in such a way where proper words are not being used so much anymore.
- Slipshod: of poor quality, not well-crafted (-)
 - Maria did not spend much time on her science project, so that is why it turned out slipshod.
- Flippantly: talking about something jokingly but that deserves respect, rude, disrespectful (-)

Review from last Study Sheet

Transition words:

- These words will mark a difference in content between two clauses of a sentence:
 - In spite of
 - But
 - Despite
 - Despite receiving the award, Heather was still not fully satisfied with her acting performance.
 - However
 - James got a B in English; however, the English teacher still felt he was a stellar student.
 - Although
 - Even though
 - In contrast
 - Contrarily

- These words will mark a similarity or emphasis between two clauses:
 - In fact
 - Percy's lemonade sales are doing great; in fact, he made over \$3,000 last month.
 - What's more
 - Moreover
 - Additionally
 - In addition

- These words indicate a conclusion is to follow from prior premises:
 - Thus
 - Tom's fingerprints were the only ones at the crime scene; thus, the detective concluded that he committed the theft.
 - Therefore
 - So
 - In light of the foregoing
 - Hence
 - Ergo